

KSHITIZ TIWARI

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SUMMARY

Ph.D. candidate in Computer Science specializing in post-training and alignment of large language models, with a focus on controllable text generation and causal modeling. Experience designing multi-objective optimization frameworks, reward-guided generation, and evaluation pipelines for aligning LLM behavior. Published researcher (Entropy 2024, ACL-IJCNLP 2022) with expertise in LoRA/QLoRA, reinforcement learning for LLM alignment (PPO/DPO), and scalable evaluation of multi-objective generation systems.

EDUCATION

Ph.D. in Computer Science, University of Arkansas

Expected Jul 2026

Research Focus: Large Language Models, Causal Inference in NLP, Robust and Fair Language Models

M.S. in Computer Science, University of Arkansas

Dec 2024

EXPERIENCE

Machine Learning Research Assistant (PhD)

Aug 2021 – Present

University of Arkansas

Fayetteville, AR

- Led research on controllable and robust NLP systems, focusing on alignment, adversarial robustness, and post-training of large language models.
- **Robust Hate Speech Detection (ACL-IJCNLP 2022)**: Developed a BERT-based model with causal regularization, achieving 91.6% accuracy and improving robustness by +23% (word substitution) and +18% (misspellings), with 0.92+ recall.
- **Adversarial Detection via MDL (Entropy, 2024)**: Proposed an information-theoretic framework using masked language modeling to detect adversarial perturbations without access to original inputs, establishing a principled robustness signal based on description length shifts.
- **Counterfactual Generation with LLMs (Under Review)**: Designed alignment-oriented training objectives for controllable text generation using geometric representation shifts without parallel data, improving style accuracy from 48% to 78% while preserving semantic similarity up to 78%.
- Investigated reinforcement learning and preference optimization (PPO / DPO) with reward models based on auxiliary BERT classifiers to steer LLM outputs toward target attributes.
- Built scalable evaluation pipelines combining embedding similarity (SBERT), BERTScore, classifier-based metrics, and LLM-as-a-judge frameworks to systematically evaluate alignment trade-offs.

PUBLICATIONS

- Tiwari, K.; Zhang, L. *Geometric Control of Language Models for Counterfactual Text Generation*. (Manuscript Under Review).
- Tiwari, K.; Zhang, L. *Implications of Minimum Description Length for Adversarial Attack in NLP*. Entropy, 2024.
- Tiwari, K.; Yuan, S.; Zhang, L. *Robust Hate Speech Detection via Mitigating Spurious Correlations*. ACL-IJCNLP, 2022.

SELECTED PROJECTS

LLM-Based Resume Screening and Candidate Ranking: Built an end-to-end candidate ranking pipeline using LLM-based resume parsing and SBERT embeddings for semantic retrieval. Designed composite ranking metrics for top-K selection, improving relevance of candidate-job matching.

Multimodal Hate Meme Detection: Developed a multimodal classification system combining Vision Transformers and BERT for joint image-text reasoning, improving robustness over unimodal baselines on hate meme detection tasks.

Semantic Search Benchmarking: Evaluated classical (LSA, LDA) and transformer-based (SBERT) retrieval methods, demonstrating superior semantic matching performance of embedding-based approaches over traditional baselines.

SKILLS

Programming: Python, C++, SQL

LLM & NLP: PyTorch, Hugging Face Transformers, PEFT (LoRA/QLoRA), TRL (RLHF/PPO/DPO), Sentence-Transformers, BERT, SBERT

RAG & Retrieval: Dense Retrieval, Embedding Models (E5, SBERT), Vector Databases (Qdrant), Semantic Search, Chunking Pipelines

Scaling & Training: Accelerate, DeepSpeed, bitsandbytes

Evaluation & Experimentation: BERTScore, Embedding Similarity, Ranking Metrics (Oracle@K), LLM-as-a-Judge, Weights & Biases (W&B)

ML & Data: Scikit-learn, XGBoost, Pandas, NumPy, SpaCy, NLTK